

# ConstructionSceneReady: Evidence-grounded compilation and task-scene validation for BIM-driven robotic assembly

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## Abstract

Robot-ready construction simulation requires more than converting building geometry into a simulation format. We ask which building information actually supports a robotic construction task in simulation, and how its absence, error, or imprecision changes task outcomes. ConstructionSceneReady compiles IFC source into task-activated multi-scale OpenUSD scenes and validates them against an explicit task-scene contract with evidence-bounded repair. On 105 fault-injection cases the contract recovers all 141 findings without false positives and repairs 97.1% to pristine equivalence; an agent constrained to the same whitelist preserves it, while an unconstrained agent produces incorrect-but-passing scenes in 22.7%. Paired humanoid episodes show symbolic and referential defects are discretely fatal and clearance is the most sensitive continuous property. The 12mm reward criterion also masks anchor-relative interface errors: geometrically valid insertion collapses to 2.0% at a 10mm hole-coordinate error while reward success stays  $\sim 80\%$ , a blind spot the contract closes with two independently verified rules.

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**Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.